# Helios Compute Control - illustrative venture investment committee memorandum

SYNTHETIC – NOT INVESTMENT ADVICE

**Conditional posture:** `INVEST` subject to the executable terms and falsifiers below.

**Authority:** `DECISION_RECORD_INCOMPLETE` / `PENDING_FOUNDER_SIGNATURE` / no delegated investment authority.

**Knowledge cutoff:** `2026-08-29T00:00:00Z`

**Packet receipt:** `5ea2a76fa53499c6e8eb9f10cc4b60c84afa7d4aae0f5b07a98244149350b8cf`

## Recommendation and executable terms

Invest only if the 30% XIRR, 3.0x MOIC, and 10% loss hurdles clear and milestone funding stays tied to retention, pipeline, and margin evidence.

| TERM | EXACT SELECTED-CASE POSITION |
| --- | --- |
| Security | Series C, 1x non-participating preference, senior to A/B and junior to any declared bridge |
| Initial / conditional capital | $25M at close / $15M at month 12 |
| Valuation | $160M pre-money / $200M fully funded post-money |
| Series C ownership | 13.51% at first close / 20.00% fully funded |
| Option pool | 1,611,031 unissued shares after refresh; pre-money holders bear the refresh |
| Selected milestone returns | 60.26% gross XIRR / 9.19x gross MOIC |
| Down-round returns | 9.64% gross XIRR / 1.58x gross MOIC |
| Shortfall case | Bridge at month 36; 20.18% / 2.51x |

## Product, market, customers, competition, and business model

Helios can become the system of control for volatile enterprise GPU spend if ordinary cohorts retain and optimizer savings translate into durable platform economics.

| DIMENSION | UNDERWRITING JUDGMENT | DISCONFIRMING FACT / OPEN GATE |
| --- | --- | --- |
| Product | GPU-spend control plane with customer-level usage and cost records. | Production replication of the optimizer effect remains open. |
| Market | Tiered Bayesian survey supports a scenario range, not a market fact. | Tier 5 is data-thin and abstained; priors remain explicit. |
| Customers | Ordinary cohorts are separated from hand-picked design partners. | Pooled retention overstates repeatability. |
| Competition | Workflow value depends on neutral provider-cost control. | Provider-native tooling and cloud concentration may compress differentiation. |
| Business model | Usage-linked expansion can compound with customer compute spend. | Gross margin remains exposed to compute, telemetry, and support costs. |
| Terms | Milestone financing limits capital at risk before operating proof. | Failure creates explicit down-round or bridge dilution. |

**Counterthesis:** Design-partner selection, inflated pipeline, cloud-cost exposure, and preference-heavy outcomes may make growth and TAM appear more durable than they are.

## Team and financing accountability

These are role-level evidence states, not invented biographies or reference-checked conclusions.

| DIMENSION | ASSESSMENT |
| --- | --- |
| Observable strengths | CEO / founder: Product insight is supported by design-partner problem definition. CTO: Optimizer experiment shows synthetic unit-cost improvement. CRO / revenue: Pipeline stage history exists at opportunity level. Finance: No dedicated capability is evidenced. |
| Unproven capabilities | CEO / founder: Commercial dependence and succession remain open. Evidence state: OPEN. CTO: Provider concentration and production replication remain open. Evidence state: PARTIAL. CRO / revenue: Ordinary-customer repeatability is not established. Evidence state: OPEN. Finance: Usage margin, runway, and financing controls need an accountable owner. Evidence state: ABSENT. |
| Key-person risk | OPEN - founder commercial dependence and succession evidence remain conditions to the second tranche. |
| Required capacity | Finance leader accountable for usage margin, monthly close, runway, and financing controls. Revenue operations owner accountable for ordinary-customer stage-to-close governance. |

## Cap table and financing-event bridge

| SCENARIO | FUNDED SERIES C | OWNERSHIP | MINIMUM CASH | EXIT PROCEEDS | GROSS XIRR | GROSS MOIC |
|---|---|---|---|---|---|---|
| BASE | $25M | 12.94% | $13.4M | $181.1M | 48.55% | 7.24x |
| MILESTONE | $40M | 20.00% | $17M | $367.7M | 60.26% | 9.19x |
| DOWNSIDE | $25M | 10.14% | $9M | $39.6M | 9.64% | 1.58x |
| FINANCING_SHORTFALL | $25M | 10.79% | $250,000 | $62.7M | 20.18% | 2.51x |

| SELECTED EVENT | MONTH | STATUS | CASH | NEW SHARES | FULLY DILUTED SHARES |
|---|---|---|---|---|---|
| series-c-close | 1 | FUNDED | $25M | 1,814,223 | 13,425,254 |
| series-c-tranche | 12 | FUNDED | $15M | 1,088,534 | 14,513,788 |

## Milestone financing and monthly runway

The $15M second tranche releases only when `ALL_TESTS_PASS_AFTER_CURE` after a 30-day cure. Evaluator: `BOARD_FINANCE_COMMITTEE`. Failure consequence: `WITHHOLD_TRANCHE_AND_REUNDERWRITE_RUNWAY`.

- `hx-nrr-metric` >= `1.05` over `trailing_12_months`.
- `hx-margin-metric` >= `0.70` over `trailing_3_months`.
- `hx-pipeline-audit` == `COMPLETE` over `as_of_test`.
- `hx-optimizer-replication` == `REPLICATED` over `pre_tranche`.

| YEAR | ENDING CASH |
|---|---|
| 1 | $48.2M |
| 2 | $38.5M |
| 3 | $30.2M |
| 4 | $23M |
| 5 | $17M |

## Preference waterfall and investor return bridge

The selected exit equity value is not a naked outcome assumption. It is derived from the retained LTM revenue ledger and a declared five-year operating and valuation scenario:

| EXIT-VALUE OPERAND | SELECTED MILESTONE CASE | CLASSIFICATION |
|---|---|---|
| Observed LTM revenue | $57M | Retained synthetic P&L |
| Annual revenue growth / hold | 48.00% / 5 years | Scenario |
| Terminal revenue / exit multiple | $404.4M / 4.00x | Scenario |
| Exit enterprise value / cash | $1,617.6M / $17M | Accounting bridge; net debt is -$17M |
| Exit equity value | $1,634.6M | Waterfall operand |

Exit value basis is `EQUITY_VALUE`. Unissued pool shares receive zero proceeds. Class proceeds conserve to $1,634.6M with a 0-cent residual.

| CLASS | ELECTION | PREFERENCE | RESIDUAL | TOTAL PROCEEDS |
|---|---|---|---|---|
| SERIES_A | CONVERT | $0 | $259.7M | $259.7M |
| SERIES_B | CONVERT | $0 | $234.4M | $234.4M |
| SERIES_C | CONVERT | $0 | $367.7M | $367.7M |

Series C invests $40M in dated funded tranches and receives $367.7M at exit: **60.26% gross XIRR / 9.19x gross MOIC**.

## Conditional distribution and sensitivity

The 1,000 retained paths replay the full financing ledger and exact waterfall. They are scenario priors, not a forecast or evidence of real-world accuracy.

Declared scenario-state priors: Base 30.00%; Downside 15.00%; Financing Shortfall 10.00%; Milestone 45.00%.

| STATISTIC | P10 | P50 | P90 |
|---|---|---|---|
| Gross MOIC | 1.54x | 7.05x | 11.72x |
| Gross XIRR | 9.20% | 46.23% | 69.33% |

- Probability below 1.0x: **0.10%** (Monte Carlo SE **0.10 pp**, 1,000 draws).
- Sensitivity book: 11 full-engine cells across exit value, exit date, later-round price, and milestone state.

## Econometric credit and zero-credit map

- HX-06 (`CAUSAL_SYNTHETIC_ONLY`) supports only the randomized synthetic optimizer estimand and tested population.
- HX-07 (`NOT_IDENTIFIED`) receives zero base-case causal credit because adoption follows spend spikes and identification fails.
- HX-05 (`PREDICTIVE_ASSOCIATION`) is a prior-sensitive market-size scenario, not causal evidence.

## Value creation and board cadence

Combined full-model impact: **$3M minimum cash**, **$3.8M Series C proceeds**, **0.35% gross XIRR**, including an explicit **$0 interaction residual**.

| INITIATIVE | EVIDENCE CLASS | BASELINE / TARGET | OWNER | MODELED CONSEQUENCE | STOP RULE / RISK |
|---|---|---|---|---|---|
| Ordinary-cohort expansion | HUMAN_JUDGMENT | 117.07% / 125% | CRO | $75,504/month cash; $9.1M exit value; $1.5M cost | Design-partner tactics do not transfer |
| Optimizer unit economics | MIXED_CAUSAL_SYNTHETIC_AND_SCENARIO | 70.61% / 74% | CTO | $53,661/month cash; $7.7M exit value; $2M cost | Provider price changes or workload mix invalidate transferred savings |
| Enterprise sales governance | DESCRIPTIVE | 48 inflated opportunities / <15% forecast error | CRO / Finance | $0/month cash; $0 exit value; $1.2M cost | Enterprise cycle elongation remains unidentified |

### Economic mapping register

- **Ordinary-cohort expansion:** ordinary_base_arr_cents * max(0, target_nrr - observed_nrr) * gross_margin * 50% realization. Inputs: exit_multiple_on_annual_cash=10.0; gross_margin=0.7061223762551445; observed_nrr=1.1706504793654182; ordinary_base_arr_cents=3234114396; realization_factor=0.50; target_nrr=1.25
- **Optimizer unit economics:** ltm_compute_cost_cents * (1 - exp(precommitted_unadjusted_optimizer_itt_log_points)) * adoption_rate. Inputs: adoption_rate=0.65; causal_boundary=Synthetic ITT identifies the planted test population only; adoption and valuation multiple remain scenario judgments.; exit_multiple_on_annual_cash=12.0; ltm_compute_cost_cents=1138084029; multiplicative_cost_savings_rate=0.08704747623808384; precommitted_unadjusted_optimizer_itt_log_points=-0.09107139999999998
- **Enterprise sales governance:** zero_base_case_credit_until_stage_history_and_conversion_design_are_identified. Inputs: modeled_value_credit_cents=0

| PHASE | TIMING | OWNER | MILESTONE | KPI | STOP RULE |
|---|---|---|---|---|---|
| Pre-close | Before signing | Deal lead | Verify cap table, preferences, side letters, and milestone definitions. | All financing operands reconciled | Do not fund with unresolved ownership or waterfall terms. |
| Day 1 | Funding date | CEO / CFO | Lock usage-margin, runway, and ordinary-cohort definitions. | Signed operating metric dictionary | Do not release milestone capital on non-reconciled metrics. |
| Day 30 | Close + 30 days | CFO | Install weekly runway and provider-cost forecast. | Cash, committed spend, gross margin | Escalate if runway falls below financing plan. |
| Day 100 | Close + 100 days | CRO / CTO | Reconcile pipeline history and optimizer adoption economics. | Ordinary-cohort NRR, conversion, cost savings | Hold second tranche if milestones are not independently evidenced. |
| Year 1 | Quarterly through month 12 | Board | Re-underwrite ownership, runway, and exit cases. | Dilution, runway, margin, milestone state | Replan financing before the declared runway floor. |

## Falsifiers, open diligence, and limitations

- FALSIFIER: Ordinary-cohort NRR below 100%
- FALSIFIER: Pipeline conversion below 20%
- FALSIFIER: Gross margin below 65%
- FALSIFIER: Runway below 12 months post-close
- OPEN **HX-D01 · HIGH · PRE_IC** - Full opportunity stage-history export through the cutoff Owner: Revenue operations diligence lead. Decision consequence: Keep the conditional tranche withheld until conversion and weighted pipeline are recomputed from complete history.
- OPEN **HX-D02 · HIGH · PRE_SIGNING** - Design-partner contract and renewal sample Owner: Commercial diligence lead. Decision consequence: Re-underwrite ordinary-cohort transferability and remove pooled-retention credit if terms are not comparable.
- OPEN **HX-D03 · CRITICAL · PRE_TRANCHE** - Provider-level compute, telemetry, and support unit-cost ledger Owner: Technical and financial diligence leads. Decision consequence: Do not release milestone capital until gross-margin improvement reconciles to provider invoices and workload telemetry.
- OPEN **HX-D04 · CRITICAL · PRE_SIGNING** - Executed preference, pro-rata, option-pool, and side-letter schedule Owner: Deal counsel. Decision consequence: Do not fund until ownership and every waterfall scenario reconcile to executed terms.

This packet is generated from a fictional deterministic room. Synthetic causal estimates recover planted mechanisms only. Scenario outputs are not forecasts. The investment record remains unsigned and requires human IC authority.

## Receipt appendix

- MILESTONE result: `f9ff468695bb5ee3fea76121b21a4bbb590792a3987b30a5e374dd77b9d9aeb8`
- Sensitivity book: `de3b3f187e9322e5dcaff775f2e05ab119bfb413d65c3439035ba7c7f1915bbf`
- DOWNSIDE result: `c52a91b04b7c6ad14e1f92e125df355f32658d6128daaf364f4171b6a02e64fe`
- Value-creation bridge: `8996c0dd3d67eeec9d120e9a34d9188d12e95406385082ce7edd0bdbc461fc9e`